# BIO00056I Summary of main topics & core concepts

**Evolutionary Genetics & Population Genomics**

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# 1 Deterministic vs. Stochastic Evolutionary Processes

- **Definition of Evolution**: Fundamental change in allele or genotype frequencies in a population from one generation to the next.
- **Four Primary Mechanisms**: Allele frequency changes are driven by natural selection, genetic drift, mutation, and gene flow.
- **Natural Selection (Deterministic)**: Acts non-randomly on fitness differences between phenotypes, requiring phenotypic variation, a link between phenotype and fitness, and a genetic basis (heritability). Drives adaptation and typically reduces genetic variation.
- **Mutation (Stochastic)**: The ultimate random source of all new genetic variation, occurring at very low frequencies.
- **Genetic Drift (Stochastic)**: Random fluctuations in allele frequencies between generations due to sampling error, leading to a loss of genetic diversity—particularly in small populations.
- **Gene Flow / Migration**: Spatial movement between populations that reduces genetic divergence between subpopulations and restores variability.
- **Null Models**: Hardy-Weinberg equilibrium provides the baseline null model to predict expected genotype frequencies when no evolutionary forces act.

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# 2 Molecular Mechanisms of Mutation and Substitution

- **Mutation vs. Substitution**: A mutation is any newly arising genetic change in an individual; a substitution is a mutation that has successfully spread to fixation across the population.
- **Point Mutations**: Base-pair substitutions in coding regions are classified as synonymous (silent) or non-synonymous (amino acid altering).
- **Indels & Reading Frames**: Insertions and deletions (indels) within coding sequence regions can alter the reading frame via frameshift mutations.
- **Structural & Chromosomal Alterations**: Gene duplications create multigene families and pseudogenes (e.g., human $\beta$-globin gene family). Larger structural mutations include transposons, duplications, rearrangements, inversions, translocations, fission, fusion, and ploidy changes.
- **Mutational Fates**: The fate of a new mutation (fixation, loss, or balancing) is governed by selection and drift; most new neutral or deleterious mutations are rapidly lost to drift or purifying selection.

## 3 Genetic Drift and Effective Population Size ($N_e$)

- **Fixation Rules**: For a neutral allele, the probability of eventual fixation equals its current initial allele frequency ($p$), and the time to fixation is directly proportional to population size.
- **Concept of $N_e$**: Effective population size ($N_e$) represents the size of an idealized population that experiences the same rate of inbreeding or loss of genetic diversity as the real population.
- **$N_e$ vs. Census Size ($N$)**: $N_e$ is almost universally smaller than $N$ due to real-world deviations from ideal assumptions.
- **Factors Reducing $N_e$**:
  - **Reproductive Skew**: Unequal numbers of breeding females ($N_f$) and males ($N_m$) decrease $N_e$ via:

$$N_e = \frac{4N_f N_m}{N_f + N_m}$$

- **Variance in Family Size**: High variance in individual reproductive output reduces $N_e$ relative to $N$.
- **Population Fluctuations & Bottlenecks**: $N_e$ across generations is dictated by the harmonic mean:

$$\frac{1}{N_e} = \frac{1}{t}\sum \frac{1}{N_i}$$

Bottlenecks dramatically reduce $N_e$ and amplify genetic drift.

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## 4 Natural Selection: Models, Dynamics, and Modes

- **Fitness Measures**:
  - **Absolute Fitness ($W$)**: Genotype-specific rate of increase predicting absolute population numbers.
  - **Relative Fitness ($w$)**: Fitness relative to the fittest genotype ($w_{\max} = 1$).

  - **Selection Coefficient ($s$)**: Measure of the strength of selection against a genotype ($w = 1 - s$).

- **General Selection Model**: The per-generation change in allele frequency ($\Delta q$) depends on initial frequencies $(p, q)$ and relative fitness values $(w_1, w_2, w_3)$:

$$\Delta q = \frac{pq[q(w_3 - w_2) - p(w_1 - w_2)]}{\bar{w}}$$

where population mean fitness is:

$$\bar{w} = p^2 w_1 + 2pqw_2 + q^2 w_3$$

- **Dominance Effects on Selection Dynamics**: Selection against a dominant allele is rapid initially; selection against a recessive allele slows down substantially as the allele becomes rare because it is "hidden" from selection in heterozygous carriers.
- **Modes of Selection**:
  - **Directional Selection**: Favors one extreme phenotype, eliminating deleterious alleles over time without reaching a stable polymorphic equilibrium.
  - **Stabilizing Selection**: Favors intermediate phenotypes, reducing phenotypic variance.
  - **Disruptive Selection**: Favors extreme phenotypes at both ends of the spectrum over intermediate ones.
- **Frequency-Dependent Selection (FDS)**: Occurs when genotype fitness depends on its frequency in the population. Negative FDS maintains polymorphism and leads to a stable equilibrium.

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## 5 Mechanisms Maintaining Genetic Variation

- **Mutation-Selection Balance**: Rare, severely disadvantageous alleles persist at an equilibrium frequency where their rate of removal by selection equals their rate of creation by recurrent mutation ($\mu$).
  - **Dominant Deleterious Alleles**: $q \approx \frac{\mu}{s}$
  - **Recessive Deleterious Alleles**: $q \approx \sqrt{\frac{\mu}{s}}$
- **Heterozygote Advantage (Overdominance)**: Occurs when heterozygotes have higher fitness than either homozygote ($w_{AA} = 1 - s$, $w_{Aa} = 1$, $w_{aa} = 1 - t$), leading to a stable equilibrium $q = \frac{s}{s+t}$.

- **Segregational Load**: The persistent reduction in average population fitness below 1 due to continuous Mendelian emergence of lower-fitness homozygotes.
- **Genotype $\times$ Environment ($G \times E$) Interactions**: Spatial or temporal environmental heterogeneity where relative fitness rankings reverse across environments, preserving variation.
- **Genetic Correlations & Trade-offs**: Driven by pleiotropy or linkage disequilibrium, creating evolutionary trade-offs and co-adapted trait complexes.

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## 6 Population Structure and Subgroup Differentiation

- **Panmixia vs. Population Structure**: Panmixia represents random mating with no allele frequency differences between groups. Reduced gene flow creates population structure, leading to divergence in allele frequencies between subpopulations.
- **Evolutionary Drivers**:
  - **Gene Flow**: Reduces genetic differentiation between populations.
  - **Genetic Drift**: Increases differentiation between isolated populations.
  - **Selection**: Increases differentiation under divergent environmental pressures, but decreases differentiation under universally adaptive selection.
- **Fixation Index ($F_{ST}$)**: Measures genetic differentiation due to population subdivision, which reduces expected heterozygosity relative to total population expectation (Wahlund effect).

$$F_{ST} = \frac{H_T - H_S}{H_T}$$

- $H_S$: Average expected heterozygosity across subpopulations.
- $H_T$: Total expected heterozygosity calculated from mean allele frequencies across all subpopulations.
- $F_{ST} = 0$: No allele frequency differences between populations (panmixia).
- $F_{ST} = 1$: Alternate alleles fixed in different populations.
- **Isolation by Distance (IBD)**: Pattern where genetic distance ($F_{ST}$) increases continuously with geographic distance (e.g., Florida black bears).
- **Locus-Specific Selection Signals**: Loci under divergent selection show significantly elevated $F_{ST}$ compared to neutral background genome regions (e.g., wing pattern loci B/D and Yb in *Heliconius melpomene*).
- **Clustering & Multidimensional Methods**:
  - **Principal Component Analysis (PCA)**: Non-tree method to simplify and visualize high-dimensional genomic variation and genetic distance.

- **STRUCTURE Analysis**: Model-based Bayesian clustering method assigning individuals to population clusters to detect migration and admixture across physical barriers.

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## 7 Neutral Theory and the Molecular Clock

- **Neutral Theory of Molecular Evolution**: Formulated by Motoo Kimura (1983), stating that most genetic variation within and between species is selectively neutral (having little or no effect on fitness).
- **Mathematical Derivation of the Molecular Clock**:
  - In a diploid population of size $N$, with a neutral mutation rate $\mu$ per gamete per generation, there are $2N\mu$ new mutations each generation.
  - The fixation probability $u$ of any single neutral mutation by random genetic drift equals its initial frequency:

$$u = \frac{1}{2N}$$

- The substitution rate $k$ (number of fixed substitutions per unit time) is:

$$k = 2N\mu \cdot u = 2N\mu \left(\frac{1}{2N}\right) = \mu$$

- **Core Result**: The substitution rate $k$ equals the neutral mutation rate $\mu$ and is completely independent of population size $N$.
- **The Molecular Clock Concept**: Because $k = \mu$, neutral mutations become fixed at a constant rate over time, allowing sequence divergence to act as a clock.
- **Stochastic Nature**: Accumulation of substitutions is a stochastic process (contains inherent randomness).
- **Factors Influencing Clock Rates**:
  - **Mutation Rate ($\mu$)**: Varies across organisms and viral types.
  - **Purifying Selection**: Removes harmful mutations, slowing the rate of substitution. Exons experience stronger purifying selection than introns, resulting in fewer observed substitutions.

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## 8 Phylogenetic Analysis and Evolutionary Trees

- **Definition & Core Purpose**: Phylogenetics is the study of evolutionary histories among species or populations, represented via branching network graphs.
- **Homology**: Fundamental requirement that sequence or phenotypic characters must be derived from a common ancestor.
- **Tree Components & Terminology**:
  - **Nodes**: Represent internal common ancestors or speciation/split events.
  - **Branches (Edges)**: Connect nodes; length represents time or inferred evolutionary change.
  - **Tips / Leaves / Taxa**: Observed extant or sampled entities.
  - **Clade**: A monophyletic group consisting of a common ancestor and all its descendants.
  - **Topology**: The branching shape and organization.
- **Tree Types**:
  - **Cladograms**: Equal branch lengths showing common ancestry without indicating evolutionary time or change.
  - **Phylograms**: Branch lengths proportional to inferred evolutionary change (mutations per site).
  - **Time Trees**: Dated phylograms where branch lengths represent absolute dates (years or mya).
- **Tree Building Methods**:
  - **Neighbor-Joining (NJ)**: Fast distance-based method without explicit sequence change models.
  - **Maximum Likelihood (ML)**: Calculates the most likely tree topology given an explicit statistical model of sequence change.
  - **Bayesian Inference**: Iteratively calculates posterior probabilities using explicit sequence change models.
- **Tree Rooting Methods**: Outgroup rooting, midpoint rooting, and dated tips.
- **Epidemic & Evolutionary Patterns Inferred from Topology**: Demonstrates viral dynamics and lineage turnover in pathogens (e.g., Influenza A, Dengue, SARS-CoV-2).

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## 9 Conservation Genetics and Molecular Forensics

- **The Extinction Vortex**: Small population size leads to increased drift and inbreeding, causing loss of genetic diversity and inbreeding depression, which further reduces population size and increases extinction risk.

- **Inbreeding Depression**: Reduced fitness in offspring from related matings, caused by unmasking deleterious recessives and loss of overdominance.
- **Genetic Rescue**: Introduction of outside individuals into an inbred population to restore genetic diversity and eliminate inbreeding depression.
- **Evolutionary Significant Units (ESUs)**: Populations prioritized for conservation based on historical isolation, defined by reciprocal monophyly for mtDNA and significant nuclear allele frequency divergence.
- **DNA Barcoding**: Standardized species identification using specific short gene sequences (~650 bp of mitochondrial COI in animals; *rbcL* and *matK* in plants). Based on the assumption that interspecific sequence divergence is significantly greater than intraspecific variation ("barcoding gap").

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## 10 Integrative Real-World Applications & Quantitative Methods

- **Medical Genetics & Pathogen Tracking**: Mapping disease loci, quantifying pathogen substitution rates, and tracking transmission chains using time trees.
- **Agriculture & Pest Management**: Detecting selection on pesticide resistance ($F_{ST}$ outliers) and modeling crop pest adaptation.
- **Conservation Management**: Incorporating $N_e$, $F_{ST}$, ESUs, and genetic rescue to halt extinction vortices in threatened populations.
- **Genomics & Computational Tools**: Synthesis of theoretical population models ($N_e$, $F_{ST}$, $k = \mu$) with high-throughput sequencing, non-tree clustering (PCA, STRUCTURE), and phylogenetic software (MEGA, IQ-TREE, RAxML-NG).

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## 11 Population Genomics and Genome-Wide Selection Dynamics

- **Population Genomics Definition**: The expansion of population genetics from single/few loci to high-throughput, whole-genome sequencing across hundreds to thousands of individuals within a species.
- **Sequencing Technologies & Applications**:
  - **Illumina**: Primary short-read platform for population resequencing (5×–40× depth), RNA-seq, ChIP-seq, and metagenomics.
  - **Pacific Biosciences (PacBio)**: Long, accurate reads suitable for *de novo* reference assemblies.

  - **Oxford Nanopore**: Generates ultra-long read lengths; ideal for rapid field sequencing and complex structural variant resolution.
  - **Sanger Sequencing**: ABI-based lower-throughput method used for plasmid verification and targeted small-scale surveys.

- **Genome-Wide Summary Statistics**:
  - **Nucleotide Diversity ($\pi$)**: Average number of nucleotide differences per site between any two randomly chosen sequences.
  - **Segregating Sites ($S$) & Watterson's Estimator ($\theta_W$)**: Metric of diversity scaled by the number of mutating sites.
  - **Allele Frequency Metrics**: Minor Allele Frequency (MAF) and Derived Allele Frequency (DAF).

- **Linkage Disequilibrium (LD) & Sweep Signatures**:
  - **LD Decay**: Non-random association of alleles at distinct loci, broken down over time via recombination.
  - **Selective Sweeps**: Positive selection drives a beneficial mutation to high frequency, sweeping adjacent linked neutral variants along with it.
  - **Signatures**: Localized drop in nucleotide diversity ($\pi$), marked increase in LD, excess of rare alleles, and high Extended Haplotype Homozygosity (EHH).

#### 11.0.1 Genomic Patterns Across Modes of Selection

| Selection Mode | Diversity ($\pi$, $\theta_W$) Effect | Allele Frequency Spectrum | Haplotype Structure |
|---|---|---|---|
| **Purifying (Negative)** | Reduced in functional regions (exons, promoters) | Excess of rare, low-frequency alleles (negative Tajima's $D$) | Baseline decay |
| **Adaptive (Positive)** | Locally eliminated surrounding the swept allele | Rapidly fixed/high DAF for beneficial allele | Long, conserved haplotypes (high EHH / $nS_L$) |
| **Balancing** | Elevated surrounding target locus | Excess of intermediate-frequency common alleles | Divergent persistent haplotypes |

- **Polygenic Adaptation & Distribution of Fitness Effects (DFE)**:
  - **DFE Profile**: Most new mutations are neutral or slightly deleterious; strongly advantageous mutations ($s > 0$) are extremely rare.

- **Polygenic Architecture**: Most complex traits adapt through subtle, simultaneous frequency shifts across many loci.

- **Population Genomic Case Studies**:
  - **Drug Resistance Sweeps**: Pyrimethamine selection on *dhfr* in *P. falciparum*.
  - **Haplotype-Based Selection Tests**: $nS_L$ statistic applied to *P. vivax* targets like the $AP2$ transcription factor.

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## 12 Comparative Genomics, Evolutionary Rates, and Divergence

- **Diversity vs. Divergence**:
  - **Diversity**: Polymorphic variation present within a species population.
  - **Divergence**: Fixed sequence differences accumulated between isolated species over time.

- **Comparative Genomics Pipeline**: Assembly, annotation/homology search, synteny analysis, and multi-species reference alignments.
- **Determinants of Evolutionary Rates**:
  - **Functional Essentiality**: Critical genes evolve significantly slower due to intense purifying selection.
  - **Expression Level**: Highly expressed genes exhibit lower evolutionary rates.
  - **Protein Structural Constraints**: Core residues evolve slowly; surface-exposed residues evolve rapidly.

- **Quantifying Evolutionary Constraint**:
  - Conserved non-coding regions and coding exons show reduced divergence across multi-species alignments due to purifying selection.

- **Sequence Divergence Tests for Selection**:
  - **Synonymous Rate ($dS$ or $Ks$)**: Amino acid-preserving substitutions; selectively neutral baseline.
  - **Nonsynonymous Rate ($dN$ or $Ka$)**: Amino acid-altering substitutions; subject to functional selection.
  - $\frac{dN}{dS}$ **Ratio Interpretation**:
    - $\frac{dN}{dS} < 1$: Purifying selection.
    - $\frac{dN}{dS} = 1$: Neutral sequence divergence.
    - $\frac{dN}{dS} > 1$: Adaptive positive selection.

- **The McDonald-Kreitman (MK) Test**:

- Compares nonsynonymous-to-synonymous polymorphism within species ($P_n/P_s$) against divergence between species ($D_n/D_s$).
- **Adaptive Divergence**: $D_n/D_s > P_n/P_s$.
- **Balancing Selection**: $P_n/P_s > D_n/D_s$.

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## 13 Continuous Variation, Phenotypic Components, and Heritability

- **Continuous vs. Discontinuous Traits**: Continuous quantitative traits are determined by multiple genetic loci interacting with environmental factors.
- **Components of Phenotypic Variance ($V_P$)**:

$$V_P = V_G + V_E + V_{G\times E}$$

- **Components of Genetic Variance ($V_G$)**:

$$V_G = V_A + V_D + V_I$$

- **Additive Genetic Variance ($V_A$)**: Cumulative phenotypic effects of individual allele substitutions (primary basis for parent-offspring resemblance).
- **Dominance Genetic Variance ($V_D$)**: Results from allelic interactions at the same locus.
- **Epistasis Genetic Variance ($V_I$)**: Arises from inter-locus interactions.
- **Broad-Sense vs. Narrow-Sense Heritability**:
  - **Broad-Sense Heritability ($H^2$)**: $H^2 = \frac{V_G}{V_P}$
  - **Narrow-Sense Heritability ($h^2$)**: $h^2 = \frac{V_A}{V_P}$
- **Estimating $h^2$ via Regressions**:
  - Mid-parent regression slope $b$ gives $h^2 = b$.
  - Single-parent regression slope $b$ gives $h^2 = 2b$.
- **The Breeder's Equation**:

$$R = h^2 S$$

- Selection Differential ($S = M_S - M$) and Response to Selection ($R = M' - M$).

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# 14 Quantitative Trait Loci (QTL) Mapping and GWAS

- **Quantitative Trait Loci (QTL)**: Specific chromosomal regions statistically associated with variation in a quantitative trait.
- **Bi-Parental Mapping Populations**: $F_2$, Backcross ($BC$), Recombinant Inbred Lines ($RILs$), and Near Isogenic Lines ($NILs$).
- **QTL Mapping Methods**: Single Marker Analysis ($ANOVA$), Interval Mapping ($IM$), and Composite Interval Mapping ($CIM$).
- **Genome-Wide Association Studies (GWAS)**: Scans genome-wide SNPs across natural populations to detect marker-trait associations using historical recombination.
- **Linkage vs. Linkage Disequilibrium (LD)**: Linkage refers to physical co-localization on a chromosome; LD represents statistical non-random association of alleles across a population.
- **Factors Governing LD Decay**: Recombination rate, population size ($N_e$), mating system (selfing vs. outcrossing), natural selection, and population structure.
- **Methodological Controls & Missing Heritability**: Mixed linear models and PCA correct for population structure. Missing heritability describes the gap between pedigree $h^2$ and GWAS SNP-based heritability ($h^2_{\text{SNP}}$).

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# 15 Genome Evolution and Structural Dynamics

- **Definition and Complexity (The C-Value Paradox)**:
  - **Genome Size vs. Complexity**: Prokaryote genome size correlates linearly with complexity. Eukaryotes show no correlation between genome size and morphological complexity (**C-value paradox**).
  - **Diversity**: Ranges from small eukaryotic genomes to massive genomes (*Homo sapiens* 3.3 Gbp, *Ambystoma mexicanum* 32 Gbp, *Paris japonica* 149 Gbp).

### 15.0.1 Structural Organization: Prokaryotes vs. Eukaryotes

| Feature | Prokaryotic Genomes | Eukaryotic Genomes |
|---|---|---|
| **Chromosome Structure** | Typically single circular chromosome | Multiple linear chromosomes |
| **DNA Packaging** | Condensed in nucleoid via supercoiling | Condensed in nucleus via histone complexes |

| Feature | Prokaryotic Genomes | Eukaryotic Genomes |
|---|---|---|
| **Ploidy & Gene Copy** | Haploid (single gene copy) | Mostly diploid (two copies per gene) |
| **Expression Coupling** | Simultaneous transcription and translation | Spatially separated (nucleus vs. cytoplasm) |
| **Extrachromosomal Elements** | Common nonessential plasmids | Plasmids rare (mostly organellar genomes) |
| **Genome Compactness** | Highly compact; minimal noncoding DNA | Extensive non-coding introns, regulatory regions, TEs |

- **Prokaryotic Genome Evolution and Lifestyle**:
  - Free-living prokaryotes maintain larger genomes (e.g., *Sorangium cellulosum* 13 Mb); obligate symbionts undergo gene decay/streamlining (e.g., *Carsonella ruddii* 0.16 Mb).
  - **Horizontal Gene Transfer (HGT)**: Transformation, Transduction, and Conjugation.
- **Organellar Origins**:
  - Endosymbiotic theory: Mitochondria ($\alpha$-proteobacteria) and chloroplasts (cyanobacteria).
- **Eukaryotic Repetitive Elements and Transposable Elements (TEs)**:
  - **Class I (Retrotransposons)**: "Copy-and-paste" via RNA intermediate (LINEs, SINEs, LTRs).
  - **Class II (DNA Transposons)**: "Cut-and-paste" mechanism.
- **Spliceosomal Introns & Alternative Splicing**:
  - Alternative splicing expands protein diversity (e.g., *Drosophila Dscam1* generating thousands of isoforms).
  - Spliceosomal introns originated from ancestral bacterial Group II self-splicing introns.
- **Linear Chromosomes, Sex Chromosomes, & Polyploidy**:
  - Telomerase resolves end-replication problems on linear chromosomes.
  - Suppression of recombination around sex-determining loci leads to heteromorphic sex chromosomes (X/Y, Z/W).
  - **Whole Genome Duplication (WGD)**: Autopolyploidy and Allopolyploidy (e.g., *Triticum aestivum*) drive instant speciation and gene duplication.